

Investigating the Effect of Depth on Temperature and Noise Pollution

Raja Batra, Zoe Kedzierski, Claudia Nanez, Eli Rejto,
Team 8, Section 1, May 4th, 2022

Abstract— An autonomous underwater vehicle (AUV) equipped with pressure, temperature, and audio sensors was deployed into the Pacific Ocean at Dana Point, CA. The AUV was designed with the goal of determining the relationship between depth and noise pollution, as well as depth and temperature. Noise pollution is a phenomenon that can lead to increased mortality of marine organisms and as such, understanding its relationship with depth can help to locate where noise pollution is most prevalent. The depth data found from the pressure sensor was inaccurate and thus left us unable to find an explicit relationship between depth and temperature, and depth and noise pollution. Despite this, general conclusions can still be drawn from the temperature data, as we can see that temperature decreases with depth. Analysis of the microphone circuit data proved difficult as it was discovered in the post-processing period that the sample rate was too low and an adequate approach to filter out noise from the motors was not found. As the team was unable to find a relationship between depth and noise pollution due to hardware issues, it would still be valuable for future projects to investigate this topic.

I. INTRODUCTION

Marine life, mainly underwater mammals and fish, are sensitive to sound and use sound for many purposes, such as feeding, mating, and communicating. Sound travels for many kilometers underwater and unwanted noise pollution caused by human activity can disrupt sea life by preventing them from using and hearing sounds. Underwater noise pollution often occurs in areas of high human activity, such as harbors. Dana Point Harbor in Orange County, CA is home to a wide variety of marine life and it also houses many boats and other forms of water transportation. This human activity has the potential to greatly disrupt the marine life ecosystem by interfering with the natural occurring sounds.

This project aims to determine the relationship between pressure and temperature and also to determine noise pollution with respect to depth at Dana Point Harbor. To test for underwater noise pollution and for the temperature, an autonomous underwater robot will be deployed at one of the shallower places in Dana Point. Mathematical models will be run to find the expected levels of noise pollution and the expected temperature as the depth of the robot increases. These models will be compared to the experimental results collected at Dana Point in mid-April to determine the level of noise pollution.

II. EXPERIMENTAL SETUP

A. Scientific Goals

The robot was designed to collect and analyze data about the levels of noise pollution present in the coastal Southern California ecosystem.

A pressure sensor was used to determine the depth that the robot travels in the water. As the robot descends, the pressure should increase. This is because of the principle of hydrostatic pressure, where the pressure equals the force per unit area exerted by a liquid on an object. So, as the ocean depth increases, the amount of water exerted on the object increases, so the pressure increases as well. Ideally, the robot was designed to be deployed as deeply as possible to observe large amounts of change in temperature and noise levels.

The second phenomena to be observed is how the temperature of the water changes with the pressure and thus the depth of the robot. As mentioned previously, the robot moves three feet underwater. It is expected that as the depth increases, the temperature should decrease and demonstrate an inverse relationship. This is because cold water has a higher density than warm water, so cold water will sink below the warmer water that stays near the surface [1]. Therefore, the scientific goal is to observe a decrease in temperature as the robot descends and then an increase in temperature as it comes back up to the surface of the water. Therefore, it is expected that the temperature will start around the temperature of the surface and then decrease slightly before increasing again. A thermistor was used to determine the temperature of the water at various depths because they record very small changes in temperature. The resistance of the thermistor changes with the change in temperature.

The third phenomena to be observed is the level of noise pollution that exists in the shallow parts of Dana Point. It is expected that there should be moderate levels of noise pollution due to people using modes of water transportation such as kayaks and paddleboards. However, the areas tested are not deep enough for significant boat travel. To record the level of noise pollution, a microphone will be used. Using the microphone, we will be able to measure audio frequencies at various depths in hopes of finding some preliminary results on the relationship between depth and noise pollution.

B. Sensor Selection

Three unique sensors were selected to measure the necessary data required to satisfy our scientific goals. These sensors measure pressure, temperature, and audio.

To measure pressure, the team chose to use the MPX5700ASX sensor for two reasons. First, the sensor can be used to determine the depth of the robot and thus can be used to find the relationship between noise pollution and depth. Additionally, the pressure value obtained from this sensor was used to help determine the relationship between pressure and temperature. The MPX5700ASX sensor can

measure pressure values ranging from 0 to 700 kPa [2]. As the robot's path was from the surface of the ocean, to just a couple of meters deep, it was expected that the pressure sensor would be measuring values ranging from atmospheric pressure (101.325 kPa) to 121.429 kPa. Thus, the sensor was suitable for this particular path and experiment. Another reason this sensor was selected was because of its simplicity in converting from pressure to voltage. The expected output of the sensor was 0.88 V (atmospheric pressure) to 0.98 V (pressure of 121.429 kPa). This was determined by using the equation (1) which was obtained from the sensor's data sheet, where P is the measured pressure value in kPa.

$$V_{Pressure_{out}} = \frac{4.5P}{700} + 0.2 \quad (1)$$

Finally, this sensor was selected because of its fast response time, as it takes the MPX5700ASX approximately 1 ms to go from 10% to 90% of its final value when subjected to a step change in pressure [2], meaning the sensor could quickly react to changes in pressure as the robot descended on its path.

For temperature measurements, a Murata NXFT15WB473FA2B150 [3] thermistor was integrated into the circuit shown in Figure 1. This sensor was selected as it had a relatively rapid response time of 4 seconds, and provided accurate temperature data during testing. As well, the range for temperature that this sensor could record ranged from -50 degrees Celsius to 150 degrees Celsius. This range permitted for the ocean temperatures to be calculated as the Dana Point Water Temperature during April was expected to range from 14.5 - 22.5 degrees Celsius.

For acoustic measurements, the electret microphone, CME-1538-100LB was chosen for a number of reasons. This sensor is waterproof and omnidirectional which makes it ideal for taking underwater measurements. This microphone is sensitive to signals as weak as -38 dB. Finally, the frequency response curve of the CME-1538-100LB is fairly even, allowing for equal representations of most sound frequencies, as seen in Fig. 1. Finally, all of the sensors used were already in the E80 lab, which reduced the cost of the entire mission.

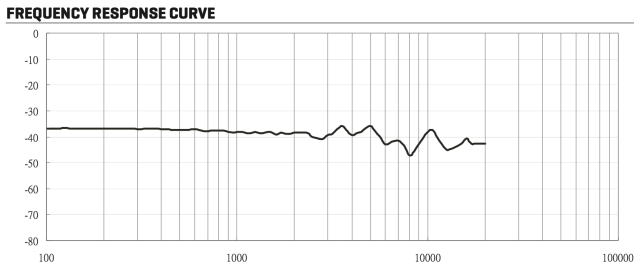


Figure 1. Frequency Response of CME-1538-100LB

C. Pressure Sensor Circuit Design

As the voltage of the pressure sensor was to be read by a Teensy Arduino board, the output needed to range from 0 to 3.3 V. While the pressure sensor alone did this, in order to get better resolution of measurements, an amplifier circuit

was designed so that the voltage would span from 1.60 V to 2.91 V for the expected path of 2 meters. This was done by designing a circuit using an MCP6002 Operational Amplifier with a gain of 12.96 V and an offset of -9.80 V, this was calculated using (2), the transfer function for the amplifier portion of the circuit (Fig. 2).

$$V_{Out} = V_{Pressure_{out}} \left(\frac{R_1}{R_3} + 1 + \frac{R_1}{R_2} \right) - \frac{5R_1}{R_2} \quad (2)$$

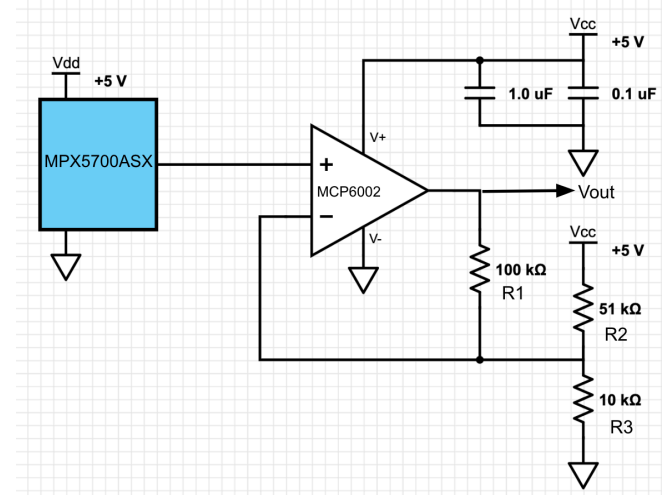


Figure 2. Pressure Circuit Schematic

D. Temperature Sensor Circuit Design

The Thermistor was put into series with a resistor at known resistance of 37 KOhms and powered by 5 volts (Fig. 3). The Voltage in between the two resistors was extracted and passed through a unity gain buffer. The OPamp used was an OP07 and the output signal Vout, was inputted into the voltage divider equation (eq. 3) to calculate the resistance of the thermistor.

$$R1 = \frac{V_{in} * R2}{V_{out}} - R2 \quad (3)$$

After the resistance of the thermistor was calculated, it was used as an input in equation 4, which was the equation given in the Murata datasheet [3]. Equation 4 converted this resistance into a temperature by solving for T.

$$R1 = 47000e^{4050 \left(\frac{1}{T+273} - \frac{1}{25+273} \right)} \quad (4)$$

The reason for selecting the 37KOhm resistor to be in series with the thermistor was because research indicated that the water in Dana Point would be around 15 degrees C in Dana Point in April. Since the teensy could only read up to 3.3 Volts, it was important that the Value of R2 did not cause the circuit to rail out in this range. As a result, R2 was selected so that 15 degrees would be the center point of Vout, which is 1.65V. Thus, using equation 2, at a temperature of 15 degrees, the thermistor has a resistance of 74.9Kohms (eq 4). Then using this value as R1 in equation 1, and a desired vout of 1.65Volts, R2 was calculated to be 37 KOhms (eq 3).

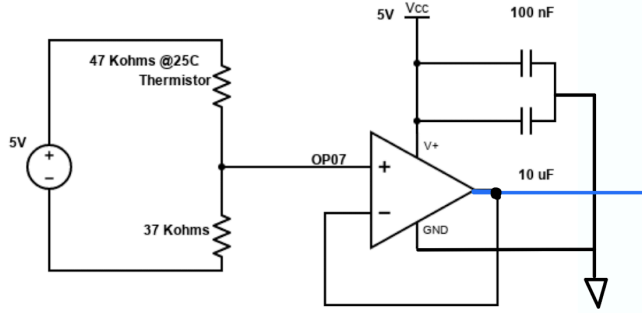


Figure 3. Schematic of Steady State Thermistor

E. Microphone Circuit Design

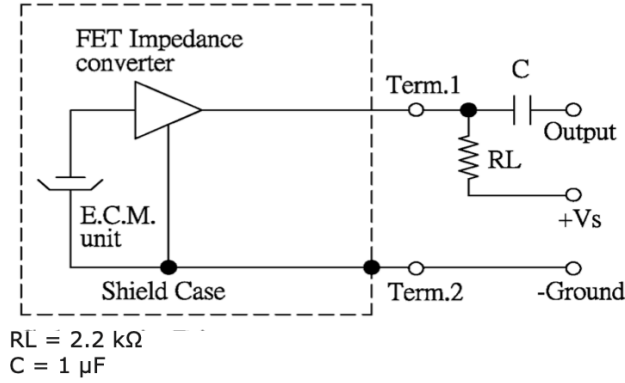


Figure 4. Recommended Schematic for use of CME-1538-100LB in a circuit

For the acoustic measurement circuit, the waterproof electret microphone, CME-1538-100LB, is recommended to be used in the circuit shown in Fig. 4. However, the output range of the above circuit is unsuitable for usage with the teensy analog inputs for multiple reasons. Primarily, the maximum output of the microphone circuit is approximately ± 20 mV, and the analog input of the teensy reads 0-3.3 V. To ensure that the analog input of the teensy was only reading positive voltage values, an offset of 1.66 V was added using a unity gain buffer shown in Fig. 5

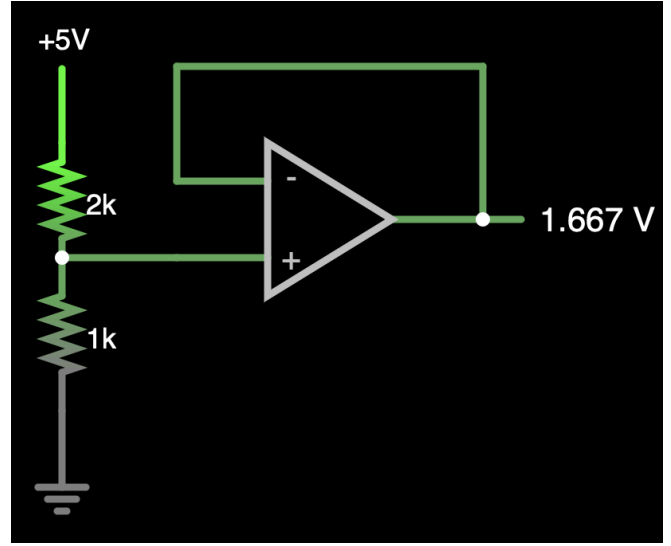


Figure 5. Unity gain buffer in microphone circuit

The second reason the raw output signal of the microphone is unsuitable for use with analog input on the teensy is because its range is so small in comparison to the 3.3 V range of the teensy analog input. For this reason, we chose to add a gain of 100 to the raw microphone output signal. To do this, two inverting amplifiers with a gain of 10 were cascaded in a manner shown in Fig. 6.

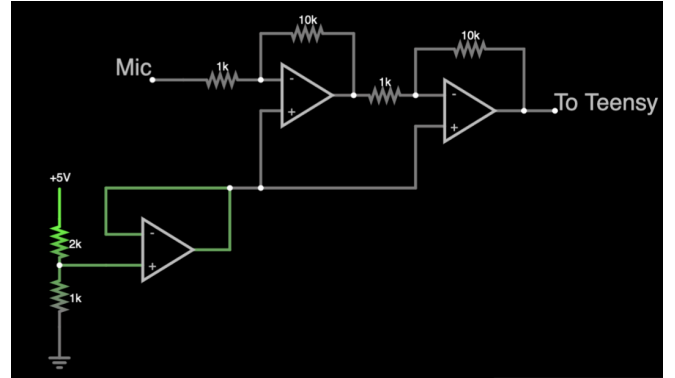


Figure 6. Final microphone circuit schematic

F. Sensor Modeling

Before deployments, expected values and models were created for each of the sensors.

In order to use the pressure sensor to obtain depth measurements so that noise pollution and temperature could be analyzed, a calibration curve (Fig. 7) was developed to convert from the pressure voltage read from the Teensy to a depth in meters. This was done in the lab by measuring the voltages at different depths in a graduated cylinder filled with water. As expected, this relationship was linear, with an R^2 value of 0.998, indicating that the sensor was working as expected. Thus, to convert from pressure to depth, (eq. 5) was used.

$$Depth = .791 * V_{Pressure_{out}} - 1.43 \quad (5)$$

Additionally, a model was created for the pressure over time of the robot's path, traveling from surface level of the ocean to two meters deep (Fig. 8). This was done by calculating the pressure at various depths using (eq. 6), where P_{atm} is atmospheric pressure, 101.325 kPa, g is the acceleration due to gravity, ρ is the density of water, 1027 kg/m^3 , and H is the depth of the robot in meters.

$$P = P_{atm} + \frac{\rho g H}{1000} \quad (6)$$

Pressure Sensor Calibration Curve

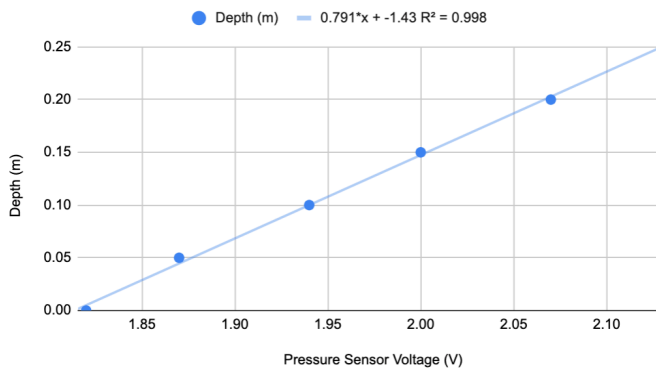


Figure 7. Pressure Sensor Voltage-Depth Calibration Curve

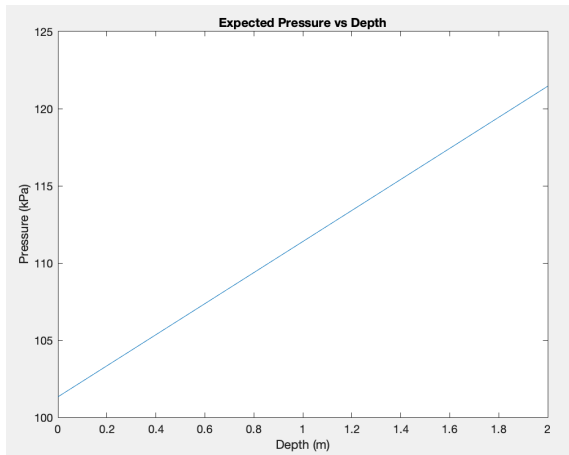


Figure 8. Expected Pressure Values During Deployment

Models were also created for the temperature sensor. The water temperature for Dana Point was predicted to be 15 degrees celsius [4], with a change of temperature of 25 degrees celsius per kilometer [5]. With our robot projected to go down 2 meters, the expected change in temperature would cause the temperature to be at 14.95 degrees Celsius. As well the expected air temperature in the morning of the launch was around 19 degrees Celsius.

G. Mechanical Design

The final robot design consisted of an AUV robot that was programmed to travel 2 meters directly down and then back up. The robot took one continuous measurement from the start of the deployment until the end when it

reached the top. Once it returned to the surface, it was collected. The final robot design being deployed is shown in Fig. 9.



Fig 9. Robot Being Deployed at Dana Point

As seen in Fig. 10, a square box with long PVC pipes extending from the top was selected for the shape. Each long leg was 30 centimeters from the edge so that the motors and the IMU would not have any interference with the magnetometer. Two motors were placed on opposite legs so that the robot could be balanced while it traveled up and down in the water. The microphone and thermistor sensors were wrapped around opposite legs and held to the PV pipe with a combination of tape and zip ties. The pressure sensor was connected to the robot to face downwards to accurately collect the pressure. The waterproof electronics box held the Teensy Arduino microcontroller and the IMU. All of the sensors and motors had wires that connected to the inside of the electronics box. These wires entered the box via penetrator bolts that were epoxied to be waterproof.

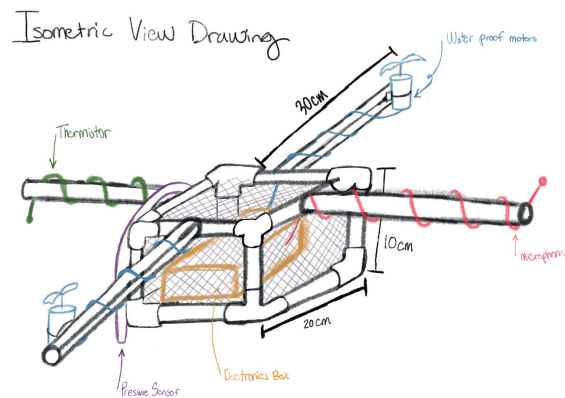


Figure 10. Isometric View of the AUV Mechanical Design

The final AUV was designed to be neutrally buoyant and designed to only have to move up and down in the water. As seen in Fig. 11, the center of mass is in the center of the robot along the x axis and roughly 2 inches down in the z axis. The total mass of the robot is 1456.8 grams and it was calculated from combining the mass of the PVC pipe, the mass of the box and the equipment inside, and the mass of the motors.

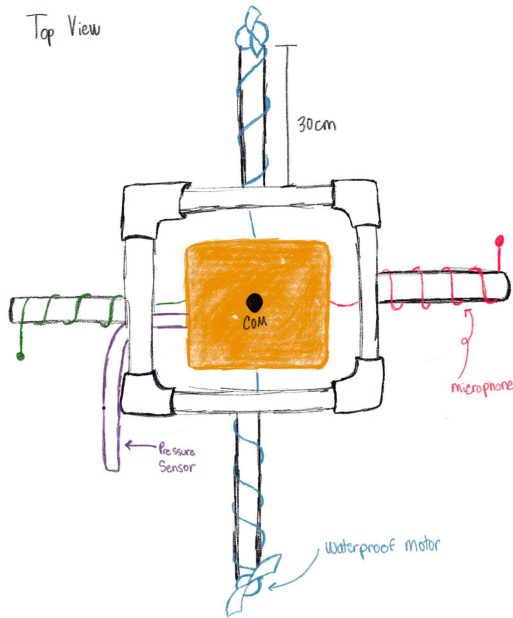


Figure 11. Top View of the AUV Mechanical Design

H. Experimental Procedure

The launch procedure at Dana Point took into account the time of data, location, and expected operation of our robot. Prior to deploying the robot, the battery voltage and current was verified. In addition, the waterproofing of the box was also tested.

The goal for launching our robot was early in the morning so that if anything went wrong, there would be time to adjust and launch again. The dock was selected to be our launch site as it provided us with enough depth for the robot to traverse several meters down to collect data.

The trajectory planned for the robot was to sit at the surface for 30 seconds, before submerging for 20 seconds and then returning back to the surface. The reason for holding the robot at the surface for 30 seconds was to gather a sample of data which would provide a standard for all the sensors.

With this plan, the robot was taken to the dock. Before submerging the robot into the water, the arduino code was uploaded to the teensy, and the motherboard along with the battery were placed and sealed in the waterproof box.

After the robot resurfaced, it was pulled out of the water and towel dried to prevent water damage. Then the payload was opened and the user switch was pressed to stop

logging. As the data was no longer being captured, the SD card was extracted from the motherboard, and uploaded so that the data could be analyzed using the Matlab scripts.

III. RESULTS AND ANALYSIS

After deploying the robot at Dana Point, on Saturday April 23, 2022, the team was able to collect data following the procedure outlined in section II.H.

A. Pressure Sensor Results

The pressure sensor was a success in some ways, but did not function entirely as expected. As seen in Fig. 12, the initial pressure that the sensor was measuring was 100.035 kPa, which is slightly under the expected atmospheric pressure value of 101.325 kPa, which is only a 1.27% error. Additionally, the general trend of increasing pressure is seen as the robot descended into the ocean (Fig. 13), however, it only increased by 1.1491 kPa, indicating a maximum depth of .114 m, (calculated using (6)) instead of the expected 2 m. Comparing this to Fig. 8, the maximum depth should have been closer to 120 kPa.

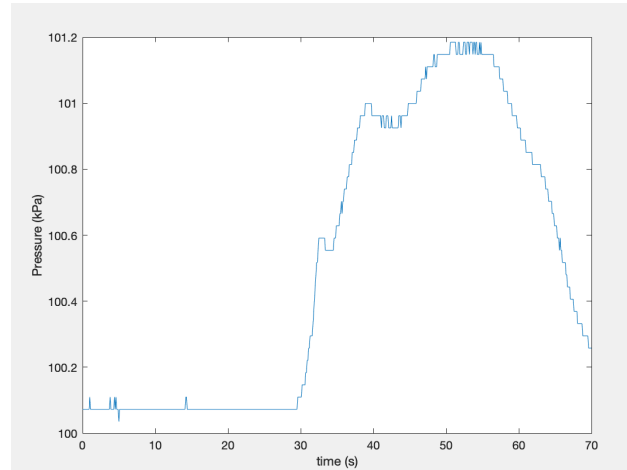


Figure 12. Pressure during the robot's deployment

Additionally, the team used the calibration curve from II.F in order to convert from pressure to depth, as seen in Fig. 13. This was done using (5). Then, the plot was shifted upwards by a constant factor so that it would indicate an initial depth of 0 m. However, after doing this, the plot (Fig. 13) showed a maximum depth of 0.08 m, which is also much lower than expected.

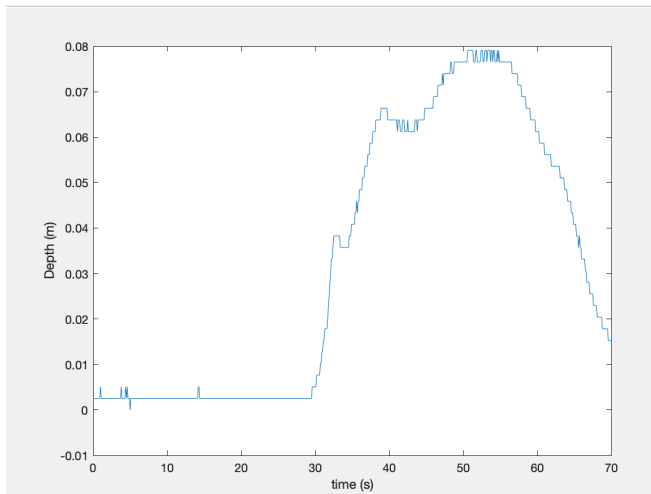


Figure 13. Calibrated depth during the robot's deployment

Given that the pressure value increases with depth, but not at the correct scale, and the linear relationship observed from the voltage and pressure discussed in section I.F, it was hypothesized that the pressure sensor functions properly, but rather there was an issue with the amplifier portion of the circuit. Future calibration and debugging is needed for this circuit.

B. Temperature Sensor Results

The data from the launch (Fig. 14) shows the temperature being at 18 degrees for the first 30 seconds. During this time, the robot was at the surface so the temperature of 18 degrees indicates the air temperature at Dana Point. 18 degrees Celsius only had an error of 2.7 percent of what the expected air temperature in Dana Point of 19 degrees Celsius. After 30 seconds, the robot began submerging. When the sensor first made contact with the water, the temperature was recorded to be at approximately 15.1 degrees celsius, which was only 0.3 percent off the predicted temperature at the surface of 15 degrees celsius. Then as the robot decreased its depth, the temperature responded as predicted and decreased as well. At the deepest point of the robot's trajectory, which was around 2 meters, the temperature decreased to approximately 14.8 degrees celsius. This was close to the expected temperature of 14.95 degrees Celsius with an error of only 0.5 percent. The change in temperatures are a few seconds delayed based on the expected trajectory of the robot as the robot began submerging at 30 seconds, but the temperature is not settled until around 33 seconds. This is a result of the time constant of the sensor.

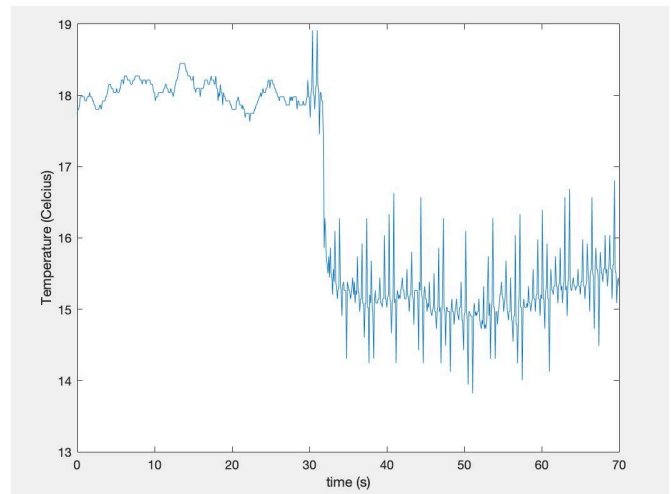


Figure 14. Temperature vs Time from launch at Dana Point

C. Microphone Circuit Results

Shown below are two graphs, one where the robot was sent underwater without the motors running (Fig. 15), and the other where the robot was sent underwater with the motors running (Fig. 16). While there are still some noises, likely from our colleagues on the dock talking and walking across the dock, there is no noticeable change in microphone output as the depth of the robot changes. This is understandable since the robot did not go very deep into the water. In comparison, when the motors are running they are the loudest part of the signal by far. Initially, the team intended to use a fast fourier transform (fft) to determine the frequencies that the dc motors emitted while running and then filter out said frequencies. Unfortunately, the sample rate of the sound data was far too low to provide enough data points for an accurate fft so this approach did not work.

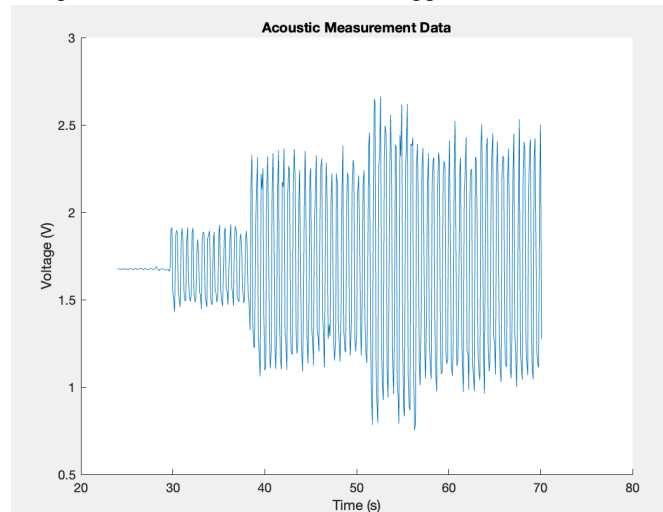


Figure 15. Microphone voltage with motors running

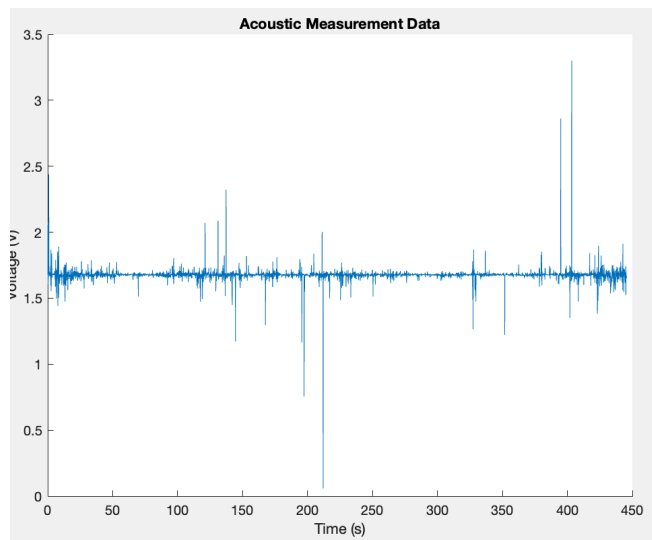


Figure 16. Microphone voltage without motors running

IV. CONCLUSION

A. Summary and Significance of Results

Overall, the results showed that our sensors were able to collect data and we could interpret data from the results. Both the thermistor and pressure sensor worked almost how they were expected to. However, the microphone data was relatively inconclusive due to the lack of sampling rate and the inability of the robot to descend further into the water.

From the temperature data, it was apparent that the temperature was dependent on the depth of the water. At around 30 seconds, when the robot began descending into the water, the pressure of the robot increased rapidly and the temperature decreased as well. At 50 seconds, the pressure peaked at its highest point of around 101 kPa. The temperature sensor experienced its lowest peak at this time as well. Therefore, it was clear that as the robot decreased in depth, the temperature also decreased in depth.

In addition, the results from the microphone indicated that the amount of noise was not dependent on the depth of the water at relatively small depths from the surface of the water. Much of the sound that was likely picked up on the no motors trial was our colleagues walking and talking on the harbor above the robot. We hypothesize that if the robot went much deeper (>20m) there would be a large enough volume of water to dampen some of the surface noise noticeably. While there was little difference between the mic signal above and below the water, there was a large discrepancy in noise between when the motors were turned off, and when the motors were turned on. While the motors were turned on, the level of noise was significantly higher, indicating that motors could be a significant source of noise pollution. This difference highlights how the many boats in the Dana Point harbor could be contributing to noise pollution near the water's surface, and thus disrupting the ecosystem.

B. Recommendations for Future Work

As mentioned in Section III.A, the team suspected an issue with the design of the amplifier portion of the pressure circuit, indicating the need for further testing and possible recalibration of the pressure sensor in order to get more accurate pressure readings. Additionally, modifying the pressure sensor so that the AUV could be deployed at greater depths without railing out in order to gather a more robust dataset for the relationship between noise pollution and depth, as well as temperature and depth is advised, as our robot only descended a couple of meters.

Additionally, it would be advisable to find a better system to filter out noise from the meters. Possible ways of doing this include a digital filter that calculates the motor's speed as it moves and then filters out those frequencies from the data. Unfortunately, the team was unable to utilize the fft algorithm to determine the frequency of the motors to filter out the sound caused by them. This was due to the low sample rate of the teensy. For sound measurements, the standard sample rate is typically 44 kHz compared to the teensy's 10 Hz. To capture more faithful audio data next time, our team could consider vastly raising the sample rate of the teensy. This would allow for an accurate fft to be performed. We could also ensure that the robot is completely neutrally buoyant and have the robot stop completely at its waypoints at specific depths, but this would require an implementation of proportional depth control as well. Finally, a hardware filter could be implemented directly onto the microphone circuit. However, the details of doing so are far beyond the scope of this project.

One other consideration for future work is modifying the mechanical design of the robot to function more smoothly. During the deployment, the robot did tip due to a lack of balance. One way to fix this would be adding motors to all of the PVC legs. Since the robot movement was omnidirectional, this would ensure that it could smoothly go up and down with minimal tilting in the right and left directions. Another way to fix the balance of the robot would be to add proper ballast and flotation to the mechanical structure.

ACKNOWLEDGMENTS

The team would like to acknowledge the E80 teaching team, Xavier Walter, Sam Abdelmuati, and Lynn Kim for all the advice and guidance they gave us on this project. We would also like to thank Stevie Steinberg for his help in debugging our microphone circuit.

REFERENCES

- [1] National Ocean Service, "Why Does the Ocean Get Colder at Depth." [online]. Available:

<https://oceanservice.noaa.gov/facts/coldocean.html>
[Accessed: 04-May-2022].

[2] “NXP Integrated Silicon Pressure Sensor On-Chip Signal Conditioned, Temperature Compensated And Calibrated MPX5700 Series.” [Online]. Available: [HTTPS://WWW.NXP.COM/DOCS/EN/DATA-SHEET/MPX5700.PDF](https://www.nxp.com/docs/en/data-sheet/MPX5700.pdf)
[Accessed: 04-May-2022].

[3] “NTC Thermistors-Murata Manufacturing.” [Online]. Available: <https://www.murata.com/~media/webrenewal/support/library/catalog/products/thermistor/ntc/r44e.ashx?la=en-us>
[Accessed: 25-March-2022]

[4] “Boneyards Surf Report & 16-day Surf Forecast - Surfline,” *Surfline*. [Online]. Available: <https://www.surfline.com/surf-report/boneyards/584204204e65fad6a7709425>. [Accessed: 04-May-2022].

[5] E. Fleming, “How does temperature change with depth?,” *sidmartinbio*. [Online]. Available: How does temperature change with depth? [Accessed: 03-May-2022].

[6] “CUI Devices Electret Condensor Mic Data Sheet” *cuidevices.com* [Online]. Available: <https://www.cuidevices.com/product/resource/cme-1538-100lb.pdf>

```
TeensyData = A02;  
Vo = (3.3/1023)*double(TeensyData);  
R1 = 100000;  
R2 = 51000;  
R3 = 10000;  
t = 0:.1:1179;  
P = ((Vo + (5*R1/R2)) / ((R1/R3) + 1 + (R1/R2))) - 0.2)*(700/4.7);  
plot(t(1),P(1))  
xlabel('time (s)');  
ylabel('Pressure (kPa)');  
  
dep = .791*Vo - 1.43;  
depth = dep + abs(min(dep));  
plot(t, depth);  
xlabel('time (s)');  
ylabel('Depth (m)');
```

CODE APPENDIX 2: Thermistor.m FUNCTION
FOR ANALYZING PRESSURE DATA COLLECTED BY THE AUV

```
TeensyData = A01;  
Vtemp = double(TeensyData)*(3.3/1023);  
R= (5*(36900)./Vtemp)-36900;  
  
for i = 1:length(R)  
    Temp(i) = 1/(log(R(i)/47000)/4050 + 1/298) - 273;  
end  
  
t = 0:.1:1179;  
plot(t,Temp)  
  
xlabel('time (s)');  
ylabel('Temperature (Celcius)');
```

CODE APPENDIX 3: MicAnal.m FUNCTION FOR ANALYZING MIC
DATA COLLECTED BY THE AUV

```
t = [1:1:(length(A03)/10)];  
doubleA03 = double(A03);  
newA03 = doubleA03.*(3.3/1023);  
  
figure(1)  
  
hold on  
plot(t,newA032)  
title('Acoustic Measurement Data')  
xlabel('Time (s)')  
ylabel('Voltage (V)')  
hold off
```

CODE APPENDIX 1: VoltageToPressure.m FUNCTION
FOR ANALYZING PRESSURE DATA COLLECTED BY THE AUV